

SAT Hard Question 12

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Question: Solving a Quadratic Equation

In the given equation, a and b are non-zero constants. Which of the following expressions gives a solution to the equation?

$$7ax^2 + (7b - 12a)x - 12b = 0$$

Choices:

A. $-\frac{a}{b}$

B. $-\frac{b}{a}$

C. $\frac{a}{b}$

D. $\frac{b}{a}$

Answer Explanation: Solving a Quadratic Equation

Correct Answer: B. $-\frac{b}{a}$

Question: Equation with Infinitely Many Solutions

In the given equation, a and b are constants. The equation has infinitely many solutions. What is the value of b ?

$$\frac{ax + b}{3} = \frac{6}{17}x - \frac{13}{12}$$

Choices:

- A. $-\frac{18}{17}$
- B. $\frac{18}{17}$
- C. $-\frac{13}{4}$
- D. $\frac{13}{4}$

Answer Explanation: Equation with Infinitely Many Solutions

For two expressions to be equal for all values of x , their corresponding coefficients must be equal.

$$\frac{ax + b}{3} = \frac{6}{17}x - \frac{13}{12}$$

Step 1: Match coefficients of x :

$$\frac{a}{3} = \frac{6}{17} \Rightarrow a = \frac{18}{17}$$

Step 2: Match constants:

$$\frac{b}{3} = -\frac{13}{12} \Rightarrow b = -\frac{13}{4}$$

Correct Answer: C. $-\frac{13}{4}$

Question: Parabola and Intercepts

In the xy -plane, a parabola has x -intercepts at $(-17, 0)$ and $(37, 0)$.
If the equation of the parabola is written as

$$y = ax^2 + bx + c,$$

where a, b, c are constants, which of the following expressions gives c in terms of a ?

Choices:

- A. $c = -1080a$
- B. $c = -629a$
- C. $c = \frac{a}{34}$
- D. $c = 51a$

Answer Explanation: Parabola and Intercepts

Solution:

Given that the parabola has x -intercepts at -17 and 37 , we can write the equation in factored form:

$$y = a(x + 17)(x - 37)$$

Expand the expression:

$$y = a(x^2 - 20x - 629)$$

Now match the standard form $y = ax^2 + bx + c$. We identify:

$$c = -629a$$

Correct Answer: B. $c = -629a$

Question: Finding the Ratio of Triangle Segment Lengths

In triangle ABC , the measure of angle B is 90° and BD is an altitude of the triangle. The length of $AB = 15$, and the length of AC is 23 greater than the length of AB .

What is the value of $\frac{BC}{BD}$?

- A. $\frac{15}{38}$
- B. $\frac{15}{23}$
- C. $\frac{23}{15}$
- D. $\frac{38}{15}$

Answer Explanation: Using Right Triangle Properties

Step-by-step:

- $AB = 15$, $AC = AB + 23 = 38$
- Use the Pythagorean Theorem to find BC :

$$AB^2 + BC^2 = AC^2 \Rightarrow 15^2 + BC^2 = 38^2$$

$$225 + BC^2 = 1444 \Rightarrow BC^2 = 1219 \Rightarrow BC = \sqrt{1219}$$

- Area using base and height:

$$\text{Area} = \frac{1}{2} \cdot AB \cdot BC = \frac{1}{2} \cdot AC \cdot BD \Rightarrow 15 \cdot BC = 38 \cdot BD \Rightarrow \frac{BC}{BD} = \frac{38}{15}$$

- **Correct Answer:**

$\frac{38}{15}$

Question: Investment Account Model

A certain investment account offers a special interest rate for the first 4 months the account is open followed by a lower interest rate for the remainder of the time the account is open.

Akeyla opened one of these accounts with an original account balance of \$700.00 and did not make any other deposits or withdrawals.

- 4 months after Akeyla opened the account, the balance had increased by 0.6% of the original balance.
- 6 months after opening, the balance had increased by an additional 0.1% of the balance at the end of the first 4 months.
- Every 2 months after the first 6 months, the balance increased by an additional 0.1% of the balance 2 months before.

Which of the following equations could represent the account balance $B(x)$, in dollars, x months after the account was opened, where $x \geq 4$?

- A. $B(x) = 704.20(1.001)^{\frac{x-4}{2}}$
- B. $B(x) = 700.00(1.001)^{\frac{x-4}{2}}$
- C. $B(x) = 704.20(1.001)^{2(x-4)}$
- D. $B(x) = 700.00(1.001)^{2(x-4)}$

Answer Explanation: Investment Growth Equation

Solution:

- Initial balance: \$700.00
- After 4 months, increased by 0.6%:

$$700 + 0.006 \cdot 700 = 700 + 4.20 = 704.20$$

- From that point on, the account grows by 0.1% every 2 months. So:

$$\text{Multiplier every 2 months} = 1.001$$

- Since the growth starts from month 4, the number of 2-month periods since month 4 is:

$$\frac{x - 4}{2}$$

- Thus, the model is:

$$B(x) = 704.20(1.001)^{\frac{x-4}{2}}$$

Correct Answer:

A

Question: Unit Conversion – Area Growth Rate

The area of a rectangular region is increasing at a rate of 180 square centimeters per hour.

What is this rate, in square meters per minute?

- A. 3
- B. 0.3
- C. 0.03
- D. 0.0003

Answer Explanation: Unit Conversion – Area Growth Rate

Step-by-step solution:

- We are given a rate of $180 \text{ cm}^2/\text{hour}$.
- First, convert square centimeters to square meters:

$$1 \text{ m}^2 = 10,000 \text{ cm}^2 \Rightarrow \frac{180}{10,000} = 0.018 \text{ m}^2/\text{hour}$$

- Then, convert hours to minutes:

$$\frac{0.018}{60} = 0.0003 \text{ m}^2/\text{min}$$

Correct Answer: D

Question: Exponential Function Evaluation

Have soem Problems

The exponential function $f(x)$ is defined by

$$f(x) = ab^x$$

where a and b are positive constants.

If **Given:**

$$f(n-1) = f(n) + \frac{82}{100}f(n-1)$$

where n is a constant, what is the value of b ?

- A. 1
- B. 0.82
- C. 0.18
- D. 10

Solve for b :

$$b \implies b = 0.18$$

Correct Answer:

Question: Exponential Function and Growth Rate

The function f is defined by the given equation, where k is a constant. The value of $f(x)$ increases by 83% for every increase of x by 1.

$$f(x) = k(1.83)^x$$

For which of the following functions, where k is a constant, does the value of $g(z)$ increase by 83% for every increase of x by $\frac{1}{4}$?

- A. $g(z) = k(1.83^x)^{\frac{1}{4}}$
- B. $g(z) = k(1.83)^{x+\frac{1}{4}}$
- C. $g(z) = k(1.83^x)^4$
- D. $g(z) = k(1.83)^{x-4}$

Question: Surface Area of Square Pyramids

A right square pyramid has a height of 11 centimeters (cm). A second right square pyramid has a height of 22 centimeters. The area of the bases of each of the two pyramids is 100 cm^2 .

Which of the following is closest to the difference in the surface area of the second pyramid and the surface area of the first pyramid, in cm^2 ?

- (A) 209
- (B) 220
- (C) 367
- (D) 551

Answer Explanation: (Part 1)

Step 1: Surface area of a square pyramid is given by:

$$\text{Surface Area} = \text{Base Area} + \text{Lateral Area}$$

The lateral area for a square pyramid is:

$$\text{Lateral Area} = 2s \cdot l$$

where s is the side of the square base and l is the slant height.

Answer Explanation: (Part 1)

Step 1: Surface area of a square pyramid is given by:

$$\text{Surface Area} = \text{Base Area} + \text{Lateral Area}$$

The lateral area for a square pyramid is:

$$\text{Lateral Area} = 2s \cdot l$$

where s is the side of the square base and l is the slant height.

Step 2: Base area is 100 cm^2 , so

$$s = \sqrt{100} = 10$$

Step 3: Use Pythagoras' theorem to find the slant height:

$$l = \sqrt{\left(\frac{10}{2}\right)^2 + h^2} = \sqrt{25 + h^2}$$

First pyramid ($h = 11$):

$$l_1 = \sqrt{25 + 121} = \sqrt{146}$$

Second pyramid ($h = 22$):

Answer Explanation: Surface Area of Square Pyramids (Part 2)

Step 4: Calculate lateral areas:

$$L_1 = 2 \cdot 10 \cdot \sqrt{146} \approx 20 \cdot 12.08 = 241.6$$

$$L_2 = 2 \cdot 10 \cdot \sqrt{509} \approx 20 \cdot 22.56 = 451.2$$

Answer Explanation: Surface Area of Square Pyramids (Part 2)

Step 4: Calculate lateral areas:

$$L_1 = 2 \cdot 10 \cdot \sqrt{146} \approx 20 \cdot 12.08 = 241.6$$

$$L_2 = 2 \cdot 10 \cdot \sqrt{509} \approx 20 \cdot 22.56 = 451.2$$

Step 5: Add base areas:

$$\text{Total Surface Area}_1 = 100 + 241.6 = 341.6$$

$$\text{Total Surface Area}_2 = 100 + 451.2 = 551.2$$

Step 6: Find the difference:

$$\Delta \text{Area} = 551.2 - 341.6 = 209.6 \approx \boxed{209}$$

Final Answer: (A) 209